15 Discovery Way, Acton, MA 01720
Phone: (978)263-3584, **Fax:** (978)263-5086
Web Site: www.piacton.com

Model ID-441
InGaAs Near Infrared Detector
with Preamplifier Operating Instructions

Ver 2.0

I. Description

The Princeton Instruments Model ID-441 InGaAs near infrared detector assembly consists of a 3mm diameter InGaAs detector mounted in a sealed housing, a preamplifier and a power supply. The preamplifier is built into the detector assembly and provides a positive voltage output proportional to the incident near IR radiation. The detector is usable over the wavelength range of 850nm to 1700nm. The power provides the necessary voltages for the preamplifier.

II. Installation

The InGaAs detector assembly mounts to the Princeton Instruments Acton monochromator exit slit using the four cap screws provided. For maximum transfer to signal, the detector should be mounted directly to the monochromator slit housing and not spaced away from the slit by a filter assembly or other accessory.

Connect the cable from the power supply module to the 9 pin connector on the detector assembly. Plug the power supply module into a 115 VAC outlet. The power supply module can remain plugged in continuously without damage to the power supply or detector.

Connect the BNC output from the detector assembly using the BNC cable provided to a readout system such as the Princeton Instruments NCL or other device capable of reading 0 to +10 Volts.

III. Operation

The InGaAs detector assembly is sensitive to near IR radiation in the wavelength range of approximately 850nm to 1700nm.

The InGaAs detector is operated in the photovoltaic mode for lowest noise performance. The preamplifier has a fixed gain to provide +10 volts out for 1.0 x 10 e-6 amps input. The detector supplied has the following specifications:

Size	mm dia.
Cd @ φ volts	_______
Rd @ 25 deg. C	_______
R [A/W] @ 1300nm	_______
R [A/W] @850nm	_______